

SAGAR-SAKSHI · EVIDENCE CARD

Smart India Hackathon 2026 | Team Tantragyan (SIH26143) | Run ID: run_kerala_20260924_074922

EVIDENCE GRADE	GRADE A
HEADLINE ASSESSMENT	Strong lead; consistent with discharge by MSC ELSA 3
AIS DATA STATUS	SYNTHETIC (Validated replay log)
MANIFEST HASH (SHA-256)	a409a25f0ce0eb0a9a230dc6f35c61c5f4667b8bbcc35491ae3fe9d8831fe70d

Candidate Hypotheses & Posterior Probabilities P(H|E)

Hypothesis / Source	Posterior P(H E)	Drift Fit	Head Prox.	Behavior
MSC ELSA 3	0.8185 (81.8%)	1.82	5.00	1.00
Null / Unknown Source (U)	0.1657 (16.6%)	N/A	N/A	N/A
DARK_sar_ship_001	0.0019 (0.2%)	0.10	0.10	1.00
DARK_sar_ship_002	0.0019 (0.2%)	0.10	0.10	1.00
DARK_sar_ship_003	0.0019 (0.2%)	0.10	0.10	1.00
DARK_sar_ship_004	0.0019 (0.2%)	0.10	0.10	1.00
DARK_sar_ship_005	0.0019 (0.2%)	0.10	0.10	1.00
DARK_sar_ship_006	0.0019 (0.2%)	0.10	0.10	1.00
DARK_sar_ship_007	0.0019 (0.2%)	0.10	0.10	1.00
DARK_sar_ship_008	0.0019 (0.2%)	0.10	0.10	1.00
OCEAN PRIDE	0.0008 (0.1%)	0.05	0.05	1.00

Exculpatory Evidence: Excluded Vessels

Vessel ID	Exclusion Reason
KERALA STAR	Track never entered backward reachability envelope (closest approach: 15.0 nm)
MALABAR QUEEN	Track never entered backward reachability envelope (closest approach: 6.9 nm)

Operational Caveats & Legal Disclaimers

- Non-Accusatory Standard:** SAGAR-SAKSHI produces investigative leads based on multi-sensor evidence consistency. It does not establish legal liability or criminal culpability.
- Detectability Mask:** SAR oil detection depends on Bragg scattering in 3-10 m/s wind regimes.
- Radar Limits:** Sentinel-1 C-band SAR resolution limits reliable vessel detection to craft >= 20m.